# Programming for Economist

PS 5 – Printing & Plotting

Class 3

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# Plan for today

1. Hints for PS5

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- Q1
  - "to the power of" in python is **, e.g. x**2
- Q2
  - A string with an f before it is called an fstring –written f'string'
  - You can put references inside an f-string via "{}"
  - You format references by {object:format}
  - Use the following format for the utility-values "8.3f"
    - This formats floats into 6 characters wide and 3 decimals
  - Enumerate returns the index and value from an array
  - Generate and print the table as you modify your code, to check f you get what you expect.

# 1. Hints for PS5

- Q3
  - To write to file use:
  - ```
    with open('file location and name', 'w') as f:
        f.write(what you want to write)
    ```
    - 'w' is for write, and f is arbitrary but normal for 'file'
    - Remember to specify file type here '.txt'
  - To read the file use:
  - ```
    with open('file location and name', 'r') as f:
        print(f.read())
    ```

# 1. Hints for PS5

- Q4
  - Use code from lectures as inspiration
  - Start by defining a function computing indifference curves(x2 as function of x1 and utility)
  - Then create an array with x1's, and looping over betas and utility you compute x2 using the function and indexing over a Boolean, to make sure points are within the constraint.
  - Useful functions
    - np.linspace
    - Use for-loops with enumerate
  - Use lecture for plot inspiration/cosmetics
- Q6
  - Use
    - `h = ax.scatter(inputs,c=u_grid)`
    - `fig.colorbar(h)`
    - To create the colour grid

# 1. Hints for PS5

- Q7
  - Similar to Q5 but instead of using the vectors you need to make a grid of possible x1's and x2's at the same time using meshgrid on the vectors.
  - Use: ax.plot_surface, and ax.contour if you don't want to compute utility yourself